

Introduction to Einstein's General Relativity

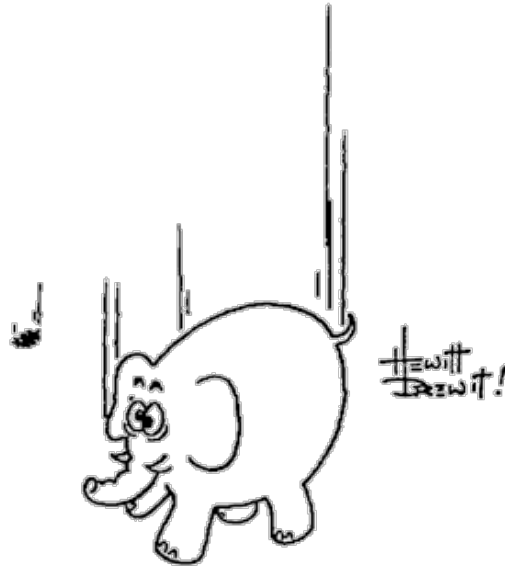
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Abstract

These are Lecture Notes for PHYS 413/513 at Rice University. At the moment, they constitute a work in progress. Please report any typos/errors to mustafa.a.amin@rice.edu.

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Preface

These are notes for a one semester course, PHYS 413/513, at Rice University. The course is meant for students (undergraduate and graduate) who have little or no prior exposure to Einstein's General Relativity. However, the students are expected to have a solid foundation in classical mechanics (with Lagrangians), multivariable calculus, differential equations, and linear algebra. Familiarity with Special Relativity and Electrodynamics will be very helpful too.

Students: Do the recommended reading before class!

Acknowledgements

The primary textbook for the course is James Hartle's "Gravity". The notes rely partly on the textbook for organization, and are very much a work in progress. Don't expect complete notes at the moment. I have added my own examples, explanations and visualizations. where I thought I had something to say which can help students further. I will also supplement these notes with exercises to be done as you read through. This will all happen in due time.

I learnt General Relativity first by reading Sean Carroll's more mathematical textbook, which I thoroughly enjoyed. I was enamoured by the elegant differential geometry at the time, but missed some of the relevant physics in that first pass. Hartle's book helped address that for me. I have also benefited from Misner, Thorne and Wheeler's "Gravitation", Weinberg's presentation in "Gravitation and Cosmology" and Wald's book "General Relativity". More recently, I have enjoyed learning from the Special and General Relativity sections of Thorne and Blandford's "Modern Classical Physics".

I owe an immense gratitude to my teachers – Prof. Eva Silverstein who first asked me to learn GR during a "rotation" during the first year of graduate school, Prof. Renata Kallosh for my first course, and Prof. Robert Wagoner for letting me TA for his GR class. This is how I learnt GR during graduate school. Prof. Barbara Shipman at the University of Texas at Arlington taught me multivariable Calculus and Linear Algebra, and then differential Geometry during my undergraduate years, both in class and often on Friday afternoons beyond class. Her enthusiasm for and pedagogy was infectious, and I hope I can follow in her footsteps.

And, importantly, the previous Phys 413/513 students at Rice were critical in making the course what it is now.

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CHAPTER 1

INVITATION

We will explore one of the most impressive achievements of our species. Along the way, I will also tell you how to lose some weight.

1.1 Going along with Gravity

Launch a coconut and a tomato with the same velocity from the earth's surface. In absence of air resistance, they will follow the same trajectory!¹ This, in itself is surprising – the trajectory is independent of their mass. If you were launched with them with the same velocity, they would simply appear to float next to you – gravity seems to have disappeared! More generally, if you were falling freely under gravity, by looking at the tomato and coconut (for a short enough period of time) they would each appear to be moving at a constant velocity, with no gravitational acceleration. Even if you, the coconut and tomato were launched with different individual initial velocities, you would still find that they appear to move at a constant velocity relative to you and each other. You will also “feel” weightless.

Exercise 1.1.1 Why is it surprising that in the gravitational field $\mathbf{g}$ the coconut and tomato, which have different masses m_c and m_t , follow the same trajectory? Here $|\mathbf{g}| = |-\nabla\Phi| \approx GM_e/R_e^2$. Consider the situation where the (approximately) constant gravitational field $\mathbf{g}$ near the earth surface is replaced by a constant electric field $\mathbf{E}$, and assume that the fruits have charges q_c and q_t . The accelerations would be $\mathbf{a}_c \propto q_c/m_c$, and similarly $\mathbf{a}_t \propto q_t/m_t$. It depends on the ratio q_i/m_i , ie. on the nature of the bodies, so the trajectories will be different! However, in the case of gravity, $\mathbf{a}_c = \mathbf{a}_t = \mathbf{g}$ independent of the mass! Check these statements using Newton's second law, as well as gravitational and electrostatic forces. See Fig. 1.1

¹Check out a version of this experiment on the moon <https://www.youtube.com/watch?v=KDp1tiUsZw8>. Also see <https://www.youtube.com/watch?v=E43-CfukEgs>.

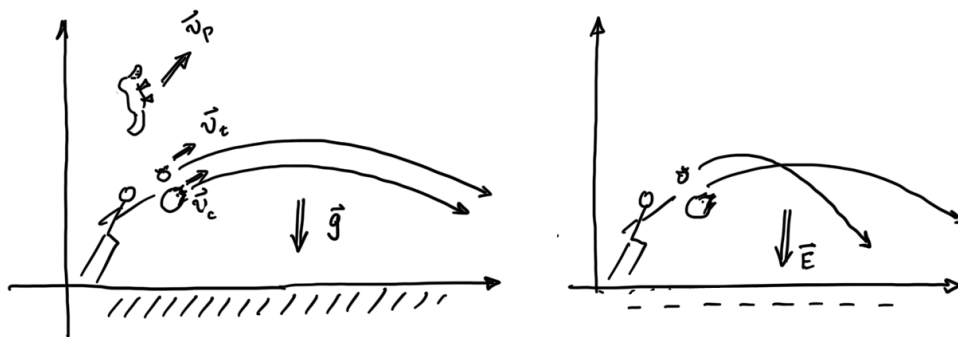


Figure 1.1

Exercise 1.1.2 Suppose that the fruit were launched from the surface of the earth with velocity $\mathbf{v}_c$ and $\mathbf{v}_t$. They will follow parabolic trajectories $\mathbf{x}_c(t)$ and $\mathbf{x}_t(t)$, with velocity $\mathbf{v}_c(t)$ and $\mathbf{v}_t(t)$ and acceleration $\mathbf{a}_c(t) = \mathbf{a}_t(t) = \mathbf{g}$. (i) Write down $\mathbf{x}_{c,t}(t)$ and $\mathbf{v}_{c,t}(t)$. Now imagine at the same time, a platypus named Polly starts falling freely from some height h , (ii) write down $\mathbf{x}_P(t)$, $\mathbf{v}_P(t)$ and $\mathbf{a}_P(t)$. (iii) What would the position, velocity and acceleration of the coconut and tomato be as measured in the “falling frame” of Polly? You should find that $\mathbf{g}$ disappears; Polly will conclude there is no gravity in the situation. (iv) Let the initial velocities $\mathbf{v}_c = \mathbf{v}_P = 0$. The coconut and Polly are falling from a height $z_c(t=0) = h_c$ and $z_P(t=0) = h_P$, both starting at rest. In this case make a sketch of t vs. $z_c(t)$ in the earth’s rest frame. Then make the same sketch in Polly’s rest frame. What do you notice? Repeat the same exercise with non-zero, and different initial velocities for Polly.

While not obvious to us on earth easily (unless you are the free, but unfortunate Polly), get in touch with your favorite “freely falling” astronaut in the international space station to understand how gravitational effects (mostly) disappear for them in their “local” laboratory on the space-station.²

²In 2024, I read a sign at the Johnson Space Center (inside the mock shuttle), which said astronauts float because of lack of earth’s gravity since they are far away from the earth. After this course, you should tell me what you think about this statement.

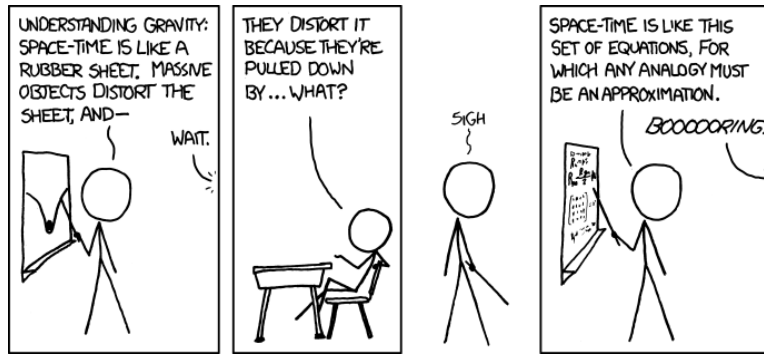


Figure 1.2: Credit: <https://xkcd.com>

With Einstein as our guide, these simple observations lead us to an astonishing insight into Nature. Gravity is unlike other known forces – it is really about the nature of space and time. Three dimensional space is not a passive arena in which events take place in time, but space and time together – *spacetime* – is an active, dynamical entity. It is a smooth collection of events: a 3+1 dimensional spacetime, whose geometry is determined by the mass-energy within it. The mass of the earth “curves” the spacetime around it.

As we will learn, despite the parabolic locus of the trajectory in space, the coconut and tomato are moving in this curved *spacetime* on the “straightest” possible path which is independent of the composition and mass of the fruit. Gravity is a manifestation of this curved spacetime.

1.2 The road ahead

We will spend much of the course developing a quantitative, calculable and testable understanding of the above ideas. Our underlying theoretical framework for gravity is that of Einstein’s General Theory of Relativity (GR). This theory, arguably one of humanity’s most elegant and well tested ideas, is built upon Einstein’s Special Theory of Relativity. We will review Einstein’s Special Theory of Relativity (SR) first – with an emphasis on geometry. The path to GR then is a matter of living with SR “locally” in small enough regions of spacetime, and developing an understanding of how to connect the happenings in many such small regions together. We will compare and contrast our developing understanding of GR with Newtonian ideas of gravity, and see why Newtonian ideas fail to be consistent with SR and more importantly, with observational and experimental tests. Nevertheless, it will be helpful to keep the following connection between Newtonian and Einstein gravity at the back of our mind: $\ddot{\mathbf{x}} = -\nabla\Phi \rightarrow$ geodesic equation, $\nabla^2\Phi = 4\pi G\rho \rightarrow$ Einstein Equations.

1.3 Newton vs. Einstein

The Newtonian theory of gravity is of course still extremely useful and accurate for most applications on the earth, solar system and in astrophysics more generally. However, Newtonian gravity fails when considering systems with mass M and extent R , such that GM/Rc^2 becomes order unity. Here $c = 3 \times 10^{10} \text{cm s}^{-1}$ is the speed of light in vacuum and $G = 6.67 \times 10^{-8} \text{dyne gm}^{-2} \text{cm}^2$

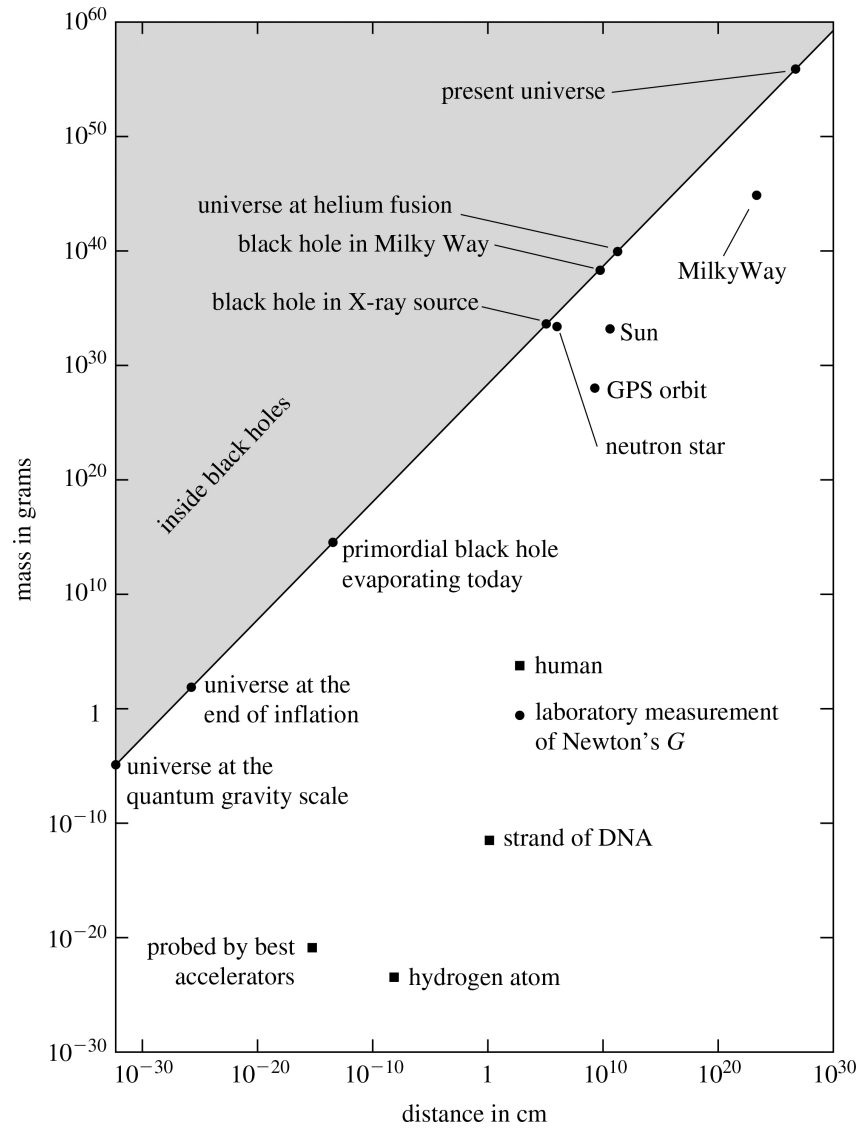


Figure 1.3: General relativity effects are not ignorable near the diagonal line, where $2GM/Rc^2 = 1$. Figure from J. Hartle. To understand the diagonal line, recall that the escape velocity from the surface of a body of mass M and radius R is $v_{\text{esc}} = \sqrt{2GM/R}$. When M and R are such that this escape velocity $v_{\text{esc}} \rightarrow c$, even light cannot escape from its surface (this is the line $M = (c^2/2G)R$). This is the historical definition of a “black hole” (Michell & Laplace, 18th century).

is Newton’s gravitational constant. Near a Black Hole, $GM/Rc^2 \sim 1$, neutron star $\sim 10^{-1}$, near the sun (say at the orbital radius of mercury) it is $\sim 10^{-8}$ and $\sim 10^{-6}$ near the sun’s surface, whereas near the earth’s surface $GM/Rc^2 \sim 10^{-9}$. Note that near the earth and the sun, the precision of the measurements is sufficiently high that even for such small values of GM/Rc^2 , we need to take deviations from Newtonian gravity into account. See Fig. 1.3 to get a feel for where different objects lie in the size-mass plane, and where how far they are from the $GM/Rc^2 \sim 1$

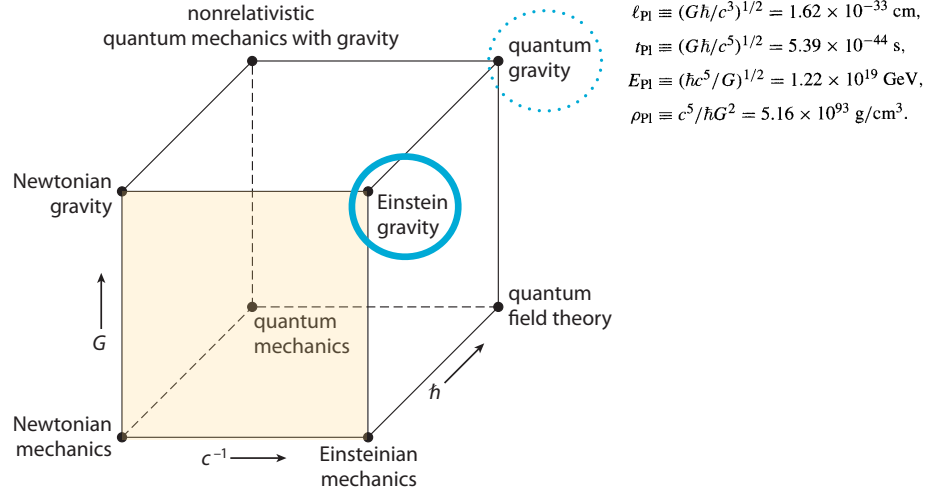


Figure 1.4: Cube of Physics, adapted from A. Zee.

region.

It is very useful to remember that for the sun $GM_{\odot}/c^2 \sim 1 \text{ km}$, where $M_{\odot} \sim 2 \times 10^{30} \text{ kg}$.

Exercise 1.3.1 Estimate $GM/(Rc^2)$ for (i) a proton (ii) human (iii) moon (iv) earth (v) sun (vi) neutron star.

1.4 What can GR do for you?

In the course, we will learn to appreciate (aspects of) the working of the Global Positioning System (GPS) relevant for your smart phones to help you orient yourself. Moving further out from our abode on our remarkable third rock from the sun, we will learn how the geometric understanding of spacetime explains the puzzling orbit of Mercury around the sun, and the observed deflection of light by the sun.³ It also describes why we can occasionally see the same supernova go off multiple times – seeing the “echoes” of the same explosion. We will be able to describe the formation and environment of spectacular ultra-compact “objects”, neutron stars and black holes in our cosmos. We will be able to understand how a dance of their binaries generate ripples in our dynamical spacetime, which have now been detected on our rock.⁴ Time-permitting we will also touch upon how the evolution of the universe as a whole relies on our understanding of the laws governing a dynamical spacetime.

³Light deflection from the sun was first observed during a total solar eclipse of 1919. For a bit of history and its significance, you might like a talk I gave on this in the context of the solar eclipse that was viewable from the continental US in 2024. <https://www.youtube.com/watch?v=D1LGSSoLfNg>.

⁴See <https://www.nobelprize.org/prizes/physics/2017/press-release/>.

1.5 Cube of Physics

To conclude, let us also put general relativity in the context of classical and quantum mechanics. A nice visual way is via the “cube of physics”, see Fig. 1.4. You should read this as, for example, when G becomes more relevant, then Newtonian gravity plays a prominent role. We will not have much to say about when quantum gravity in this course. It becomes important at the Big Bang and near the center of black holes. Notice the physical scale where quantum gravitational effects need to be taken into account. For this course, we will restrict ourselves to classical gravity.

GEOMETRY AS PHYSICS

Gravity is Geometry Considerations of universal free-fall, amongst other things, eventually led Einstein to the profound insight that gravity is not a force; it is a manifestation of the *geometry of spacetime*. He was helped by the fact that his Special Theory of Relativity (ignoring gravitational effects), initially formulated in terms of transformation laws between different observers, had already been geometrized by Minkowski.¹ This geometry, whether in Special or General Relativity, is $3 + 1$ dimensional (3 spatial and 1 temporal dimension). In General Relativity, spacetime can be intrinsically curved, and this curvature encodes gravitational effects. To build intuition for what “intrinsically curved” means, we will first study simpler examples involving two-dimensional spatial geometries.

Geometry of space (and spacetime) is something accessible to experimental verification. Gauss could in principle have tested whether the angles of a large triangle sum to π (the triangle is in three dimensions, not restricted to be on the earth’s surface).² Modern cosmological observations perform the same kind of experiment on vastly larger scales by comparing known physical scales with their observed angular sizes. You can learn about Gauss’s failed attempt to do so in the vicinity of earth. And then learn about our very successful attempt to do so in the context of cosmology. Both are described in Chapter 2 of Hartle.

2.1 Euclidean vs. intrinsically curved spatial geometry

To understand “intrinsically curved” geometry, let us first restrict ourselves to spatial geometries (as opposed to spacetime geometry). To make things easier still, let us consider 2 dimensional spatial geometries.

¹Einstein initially resisted Minkowski’s geometrization of special relativity, reportedly dismissing it as “superfluous learnedness.” He is also quoted as saying, “Since the mathematicians have invaded the theory of relativity, I do not understand it myself anymore.” The geometric viewpoint later became indispensable to Einstein in developing general relativity.

²This is historically debated a bit.

2.1.1 Two dimensions

2d Euclidean An example of an Euclidean geometry is a 2-dimensional plane, where Euclid's axioms are satisfied. For example, (i) parallel lines do not intersect, (ii) sums of interior angles of a triangle made of straight lines is π radians and (iii) the ratio of the circumference of a circle to its radius is 2π . Non-Euclidean geometry is of course a geometry where any of Euclid's axioms are violated.

2d intrinsically curved An example of an intrinsically curved 2-dimensional geometry is a 2-sphere (with radius a). Note that Euclid's axioms are not satisfied on a 2-sphere. For example: (i) Any two distinct great circles intersect, so there is no global analogue of distinct parallel straight lines on the sphere. (ii) Sums of interior angles of a triangle made of straight lines (great circles) is $> \pi$ radians ($= \pi + A/a^2$). Here, A is the area of the triangle on the 2-sphere. (iii) The ratio of the circumference of a circle to its radius is $< 2\pi$. That is, $\text{circumference}(C)/\text{radius}(r) = 2\pi \sin(r/a)/(r/a)$ where r is the radius of the circle on the 2-sphere.

Note that these properties can be tested by a bug living on the surface of the sphere, without ever leaving the surface. These are “intrinsic” properties of the surface, not requiring any information beyond the surface itself. For example, a bug could take a small segment of a string, and attach one end at some point on the sphere. Then crawl away till the string becomes taut. The bug would measure this taut length in terms of their own tiny steps. This is a radius r in units of their step size. Now, the bug can walk around keeping the string taut to trace out a circle on the sphere. Again, the circumference C can be measured in terms of the number of their tiny steps. By comparing the circumference to the radius, the bug would discover that they were living in a non-Euclidean space. They never had to see the embedded 2-sphere in 3d Euclidean space by flying out of the surface! Also note they can look for these effects without knowledge of global properties such as whether their space is finite or infinite.

A geometry is *intrinsically curved* if departures from Euclidean geometry can be detected by measurements made entirely within the space, even locally, without reference to an embedding in a higher-dimensional space. The opposite is *intrinsically flat*. Thus, a sphere is intrinsically curved, while a plane is intrinsically flat. A cylinder provides an instructive example: although it looks curved when embedded in three dimensions, it is intrinsically flat.

Intrinsically curved spatial geometries are necessarily non-Euclidean, although the term “non-Euclidean” by itself can also include global differences that do not involve intrinsic curvature.³

2.1.2 Three dimensions

3d Euclidean Our usual Euclidean 3d space is the obvious generalization of a plane from 2 to 3d.

3d intrinsically curved A 3-sphere is a 3d generalization of the 2-sphere. It is also non-Euclidean. For example the volume enclosed by a 2-sphere of radius r in this 3-sphere geometry is

$V = 4\pi a^3 \left[\frac{1}{2} \sin^{-1}(r/a) - (r/2a) \{1 - (r/a)^2\}^{1/2} \right]$. In the limit $r/a \ll 1$, this volume approaches its Euclidean counterpart $V = (4\pi r^3/3)[1 + (3/10)r^2/a^2 + \dots]$.

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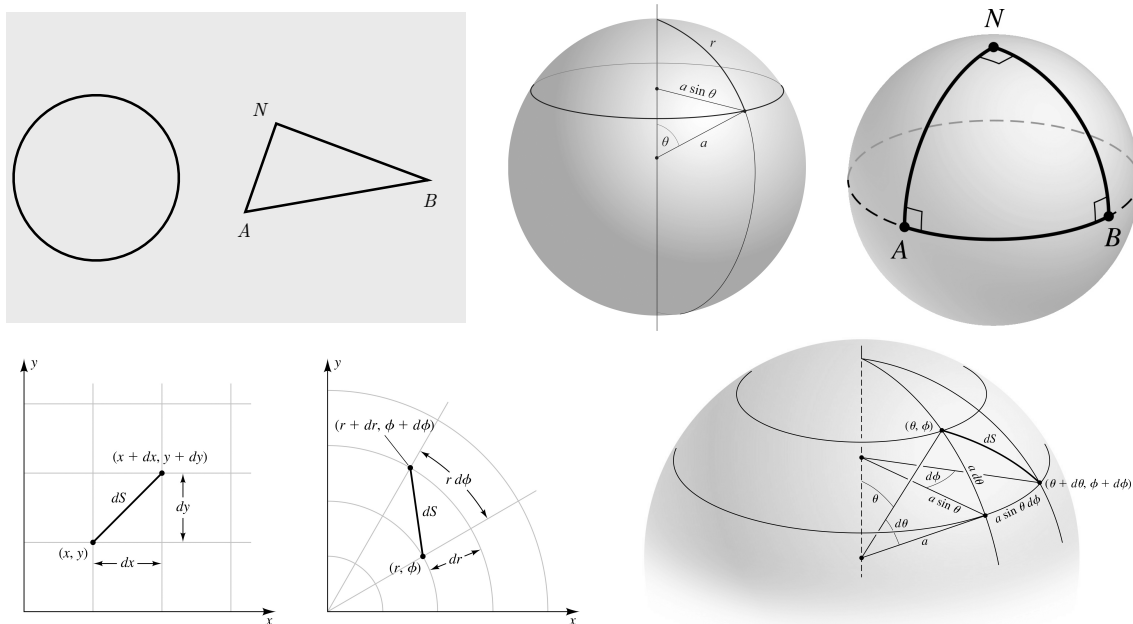


Figure 2.1: Top panel (right): The sum of the interior angles of a triangle ABN on a 2-sphere is $> \pi$, in this particular case it is $3\pi/2$. On the plane it is of course π . Top panel (middle sphere): The circumference of the circle of radius r is $< 2\pi r$, it is $C = 2\pi r \sin(r/a)/(r/a)$. On a plane it would be $C = 2\pi r$. Bottom panel (left): Cartesian and Polar co-ordinates in a 2d plane. Bottom panel (right): r and θ co-ordinates on the sphere. Part of the Fig. adapted from Hartle.

2.2 Specifying a geometry

How does one specify a geometry? A convenient and general way to do this is to define a rule for determining distances between nearby points. This is called a metric. Once this rule is specified, lengths of curves can be calculated by adding up such small distances. “Straight” lines can now be defined as curves with extremal lengths between two points. Circles (or spherical surfaces) can be defined as the locus of all points equidistant from a given point, angles can be defined as ratios of some arclength of a circle to the radius, and so on. This way of defining a geometry has significant advantages over trying to describe a geometry by embedding it in some higher dimensional Euclidean space or specifying it with a set of axioms. Both of these get cumbersome and difficult as we move to general geometries which do not have high degree of symmetry.

If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Please inform your Flat-Earther buddies.

2.2.1 Labeling and calculating with co-ordinates

Co-ordinates are labels of points in a geometry. For example, in a 2d plane, one can label points using the usual Cartesian co-ordinates (x, y) . One could also label points using polar co-ordinates

(r, θ) . Different co-ordinates can be chosen to label points in the same geometry. Different co-ordinate systems are connected by a co-ordinate transformation. For example, $x = r \cos \theta$ and $y = r \sin \theta$.

The square of the distance, dS^2 , between nearby points in a plane with labels (x, y) and $(x + dx, y + dy)$ or (r, θ) and $(r + dr, \theta + d\theta)$

$$dS^2 = dx^2 + dy^2 = dr^2 + r^2 d\theta^2. \quad (2.2.1)$$

In different co-ordinate systems the expression for the distance can look different. However, this does not mean the geometry has changed.

Exercise 2.2.1 Using $x = r \cos \theta$ and $y = r \sin \theta$, show that $dx^2 + dy^2 = dr^2 + r^2 d\theta^2$.

For a 3d Euclidean, the distance between nearby points (x, y, z) and $(x + dx, y + dy, z + dz)$ or equivalently, (r, θ, ϕ) and $(r + dr, \theta + d\theta, \phi + d\phi)$ is

$$dS^2 = dx^2 + dy^2 + dz^2 = dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.2.2)$$

The different co-ordinate systems are connected by $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$.

Similarly, for a non-Euclidean geometry like a 2-sphere of radius a , a convenient choice of co-ordinates are the latitudes and longitudes (θ, ϕ) . The distance between nearby points (θ, ϕ) and $(\theta + d\theta, \phi + d\phi)$ is

$$dS^2 = a^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.2.3)$$

For a 3-sphere of radius a , the distance between nearby points (ψ, θ, ϕ) and $(\psi + d\psi, \theta + d\theta, \phi + d\phi)$ is

$$dS^2 = a^2[d\psi^2 + \sin^2 \psi(d\theta^2 + \sin^2 \theta d\phi^2)]. \quad (2.2.4)$$

Note that this square of small distances can be conveniently written as

$$dS^2 = \sum_{i=1}^3 \sum_{j=1}^3 g_{ij} dx^i dx^j, \quad (2.2.5)$$

where $x^1 = \psi$, $x^2 = \theta$ and $x^3 = \phi$, where $g_{11} = a^2$, $g_{22} = a^2 \sin^2 \psi$, $g_{33} = a^2 \sin^2 \psi \sin^2 \theta$ with all other $g_{ij} = 0$ (the off-diagonal terms need not always be zero). We will use a simplified notation

$$dS^2 = g_{ij} dx^i dx^j, \quad (2.2.6)$$

where summation over repeated upstairs and down stairs indices is implied.⁵ Note that g_{ij} is a function of the co-ordinates $g_{ij}(x^1, x^2, x^3)$, which we will write as $g_{ij}(x^k)$ in shorthand.

For a given co-ordinate system, you can think of g_{ij} as components of a $d \times d$ matrix $\mathbf{g}$ where d is the dimension of space, with g_{ij} being the entry on the i -th row and j -th column. These entries

⁵Notice the potentially confusing notation, $dx^2 \neq (dx)^2$ in these later expressions, although earlier we were using $dx^2 = (dx)^2$ earlier.

will depend on the co-ordinate system. For example, in the case of the metric in 3d Euclidean space, the matrix entries in Cartesian co-ordinates would be $g_{ij} = \delta_{ij}$ where $\delta_{ij} = 1$ for $i = j$ and 0 for $i \neq j$. In Polar co-ordinates, $g_{11} = 1$, $g_{22} = r^2$ and $g_{33} = r^2 \sin^2 \theta$, and all other entries being 0. The components (matrix entries) look different, but together with $dx^i dx^j$ in the respective co-ordinates, they must yield the same dS^2 .

Exercise 2.2.2 Write down the components g_{ij} of the metric on a 2-sphere of radius a in terms of θ and ϕ co-ordinates.

To calculate non-infinitesimal distances along some curve γ , we can use the metric (in any co-ordinate system) as follows:

$$L[\gamma] = \int_{\gamma} dS = \int_{\lambda_1}^{\lambda_2} d\lambda \sqrt{g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}} \quad (2.2.7)$$

In the final expression, you should think of a parametrized curve $x^i(\lambda)$, and $g_{ij}(x^k(\lambda))$. The length of the curve is of course independent of the choice of co-ordinates, so it can be computed in any co-ordinate system. Again, we are using Einstein's summation convention. It might be useful sometimes to use one of the co-ordinate itself as the parameter λ .

To calculate the volume of a spatial region enclosed by a surface $\mathcal{S}$

$$\mathcal{V}[S] = \int_{\mathcal{S}} \sqrt{\det g} \prod_{i=1}^d dx^i. \quad (2.2.8)$$

where $\det g$ is the determinant of g .

Exercise 2.2.3 Use the metric on the surface of a 2-sphere to do the following: (i) Calculate the circumference a circle of radius r (distance from the center to the circumference) in the two-dimensional geometry given by the surface of a sphere of radius a . Show that this reduces to $2\pi r$ when $r \ll a$. (iii) Calculate the area of the same circle on the sphere, and again show that it reduces to πr^2 when $r \ll a$. (iv) Do the same for a cylinder of radius a and infinite extent in the perpendicular direction to the radius. Assume $r < \pi a$. Note that the cylinder is not globally Euclidean, but it is intrinsically flat.

Having hopefully gained some intuition about what intrinsically curved means, let me re-iterate that while we started with just curved space for pedagogy, ultimately we care about curved spacetime when thinking about gravitational physics.

For spacetime, the role played by Euclidean spatial geometry in the examples in this chapter, will be taken over by Minkowski spacetime geometry (which is non-Euclidean!).⁶ And, we

⁶The metric (in spatially Cartesian co-ordinates) is

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (2.2.9)$$

It is the negative sign that makes it non-Euclidean.

will define intrinsically curved spacetime geometry by their departure from Minkowski spacetime geometry.